INCH-POUND

MS27418M
w/AMENDMENT1
10 October 2024
SUPERSEDING
MS27418M
04 October 2023

MILITARY SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, 25 AMPERES, 3 PST-NO, TYPE I, HERMETICALLY SEALED, HOOK AND SCREW TERMINALS, STUD AND BRACKET MOUNTED

Inactive For New Design After 15 November 2002.
No Superseding Specification.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the latest issue of MIL-PRF-6106.

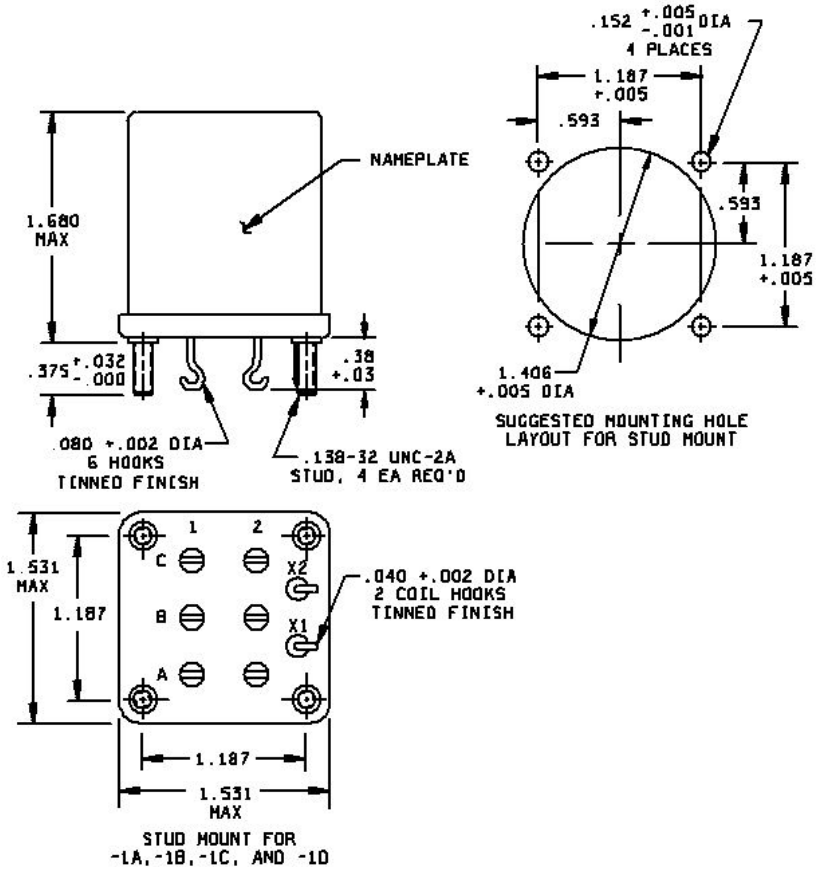


FIGURE 1. Dimensions and configuration.

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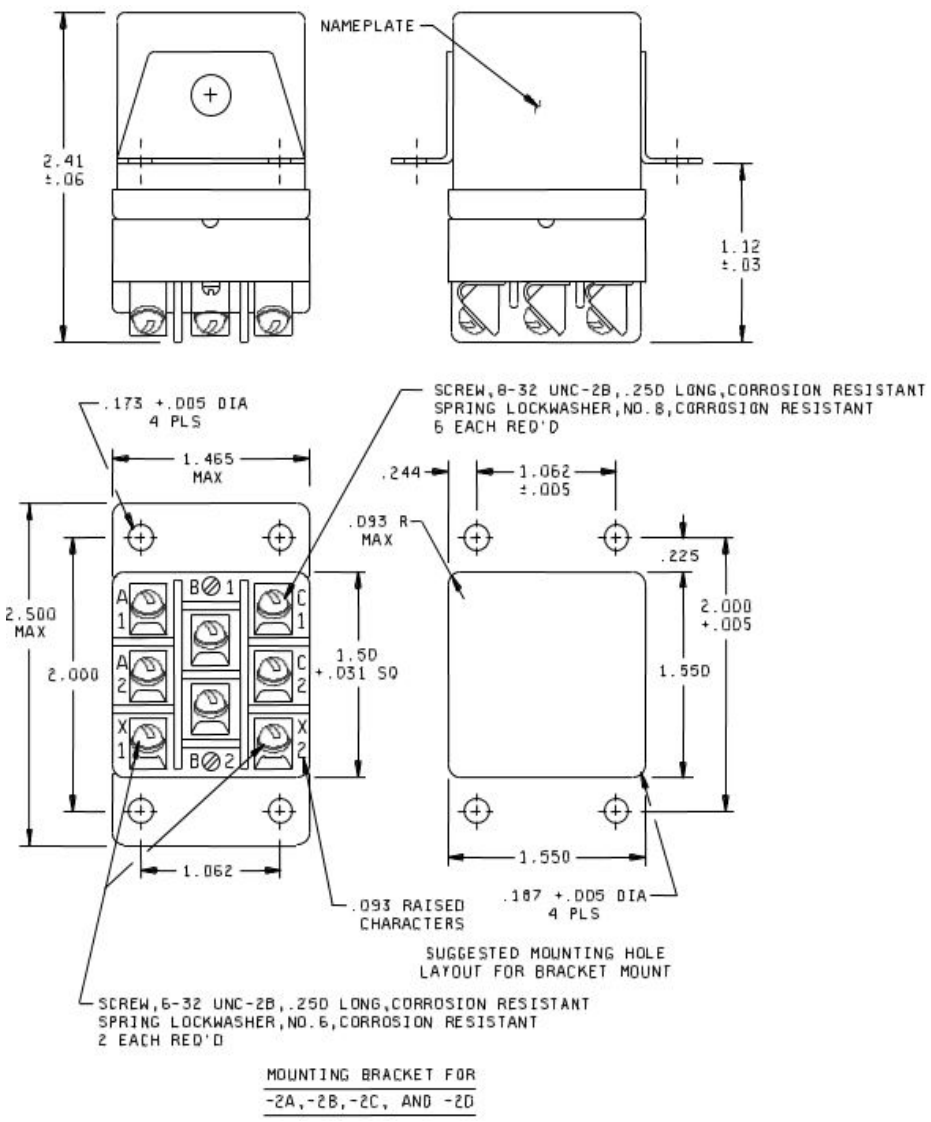


FIGURE 1. Dimensions and configurations - Continued.

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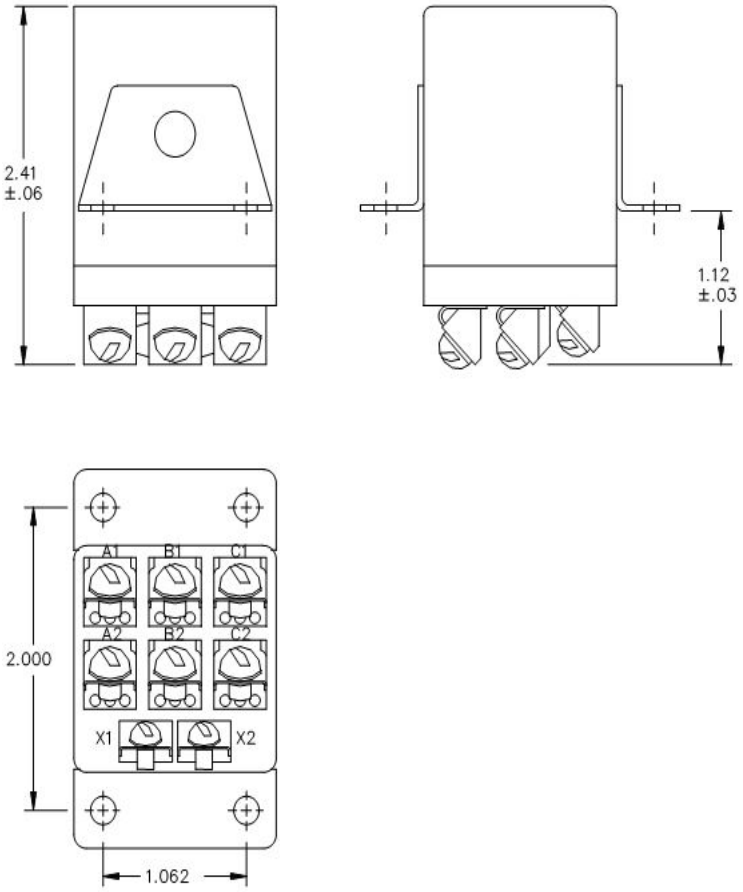
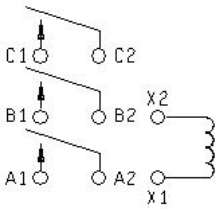
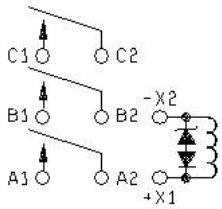


FIGURE 1. Dimensions and configurations - Continued.

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CIRCUIT DIAGRAM A
(DC UNSUPPRESSED AND AC COILS)



CIRCUIT DIAGRAM B
(DC SUPPRESSED COILS)

Inches	mm	Inches	mm	Inches	mm
.001	0.03	.138	3.51	1.12	28.5
.002	0.05	.152	3.86	1.187	30.15
.005	0.13	.173	4.39	1.406	35.71
.03	0.8	.187	4.75	1.465	37.21
.031	0.79	.225	5.72	1.50	38.1
.032	0.81	.244	6.20	1.531	38.89
.040	1.02	.375	9.52	1.550	39.37
.06	1.5	.38	9.7	1.680	42.67
.080	2.03	.593	15.06	2.000	50.80
.093	2.36	1.062	26.97	2.41	61.2
				2.500	63.50

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. A, B, and C contacts capable of accepting SAE-AS25036-53 lugs, X1 and X2 contacts capable of accepting SAE-AS25036-7 or -2 lugs, per SAE-AS25036.
5. Terminal numbers need not appear on relay headers provided there is affixed to the relay a suitable legible circuit diagram that identifies each terminal location specified.
6. In the event of a conflict between the text of this specification sheet and the references cited herein, the text of this specification sheet shall take precedence.
7. Tin/lead plating over copper plating with a minimum lead content of 3%.

FIGURE 1. Dimensions and configurations - Continued.

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TABLE I. Dash numbers and characteristics.

Dash number MS27418-	Type	Coil	Circuit	Terminal type	Mounting or mating socket	Max weight in pounds
1A	I	ac	A	Solder hook	Stud	.50
1B	I	dc	A	Solder hook	Stud	.50
1C	I	ac	A	Solder hook	Stud	.50
2A	I	ac	A	Screw	Bracket	.60
2B	I	dc	A	Screw	Bracket	.60
2C	I	ac	A	Screw	Bracket	.60
1D	I	dc	B 1/ 2/	Solder hook	Stud	.51
2D	I	dc	B 1/ 2/	Screw	Bracket	.61

1/ Transient voltage (back emf) 42 V dc maximum.
2/ Diodes shall have a peak inverse voltage of 600 V dc minimum.

TABLE II. Operating characteristics.

Part no. MS 27418-	Coil data											Time –(milliseconds maximum)					
	Coil	Nominal			Max		Max pick-up voltage			Hold volt-age 3/	Drop out volt-age 3/	Oper-ate 4/	Re-lease 5/	Contact Bounce			
		Volts 1/	Freq Hz	Res Ω ±10%	Volts	Amp	Nor-mal 3/	High temp test	Cont cur-rent test					Main		Aux	
														NO	NC	NO	NC
1A	X1,X2	115	400	N/A	122	.055	95	100	108	40	5.0	25	50	2	---	---	---
2A	X1,X2	115	400	N/A	122	.055	95	100	108	40	5.0	25	50	2	---	---	---
1B	X1,X2	28	dc	160	29 2/	.25	18	19.5	22.5	7.0	1.5	20	10	2	---	---	---
2B	X1,X2	28	dc	160	29 2/	.25	18	19.5	22.5	7.0	1.5	20	10	2	---	---	---
1C	X1,X2	115	50/ 60	N/A	122	.06	95	100	108	40	5.0	25	50	2	---	---	---
2C	X1,X2	115	50/ 60	N/A	122	.06	95	100	108	40	5.0	25	50	2	---	---	---
1D	X1,X2	28	dc	160	29 2/	.25	18	19.5	22.5	7.0	1.5	20	10	2	---	---	---
2D	X1,X2	28	dc	160	29 2/	.25	18	19.5	22.5	7.0	1.5	20	10	2	---	---	---

1/ Caution: Use of any coil voltage less than nominal coil voltage will compromise the operation of the relay.
2/ Maximum coil voltage shall be +32 V dc when maximum ambient temperature does not exceed 85°C.
3/ Over the temperature range.
4/ With nominal coil voltage.
5/ From nominal coil voltage.

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TABLE III. Rated contact load (amperes per pole) (case grounded).

Type of load	Life operating cycles x 10^3	28 V dc				115 V ac, 1 phase				115/200 V ac, 3 phase 1/				See appropriate notes
		Main		Aux		Main		Aux		Main		Aux		
		NO	NC	NO	NC	400 Hz	50/60 Hz	400 Hz	60 Hz	400 Hz	50/60 Hz	400 Hz	60 Hz	
Resistive	50	25				25	25			25	25			
Inductive	10					25	25			25	25			2/
Inductive	10	15												
Motor	50	20				20	12			20	12			
Lamp	50	10				10	10			10	10			
Transfer load														3/
Mechanical life reduced current	200	6.3				6.3	6.3			6.3	6.3			
Mixed loads	Applicable per specification													

1/ Absence of value indicates relay is not rated for 3 phase applications.
2/ 0.7 pF, inductive.
3/ Transfer load indicates relay is suitable for transfer between unsynchronized ac power supplies at rating indicated.

Environmental characteristics:

Temperature range: -70°C to +125°C.

Maximum altitude rating: 80,000 feet.

Shock g-level: 50 g's.

Duration: 11 ms.

Max duration contact opening: 2 ms.

Vibration - sinusoidal:

Operating

G-level 10 g's.

Frequency range 5 - 1,000 Hz.

Non-operation.

G-level: 15 g's.

Frequency range: 20 to 2,000 Hz.

Acceleration 10 g's.

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Electrical characteristics:

Insulation resistance, initial: 100 megohms.

After life or environmental tests: 50 megohms.

Dielectric strength (sea level):

	Initial	After life tests
Coil to case	1,250 V rms	1,000 V rms
Aux contacts	N/A	N/A
All other points	1,500 V rms	1,125 V rms

Dielectric strength (altitude):

		80,000 ft
Coil to case	N/A	500 V rms
Aux contacts	N/A	
All other points	N/A	500 V rms

Maximum contact drop initial: 0.150 volt.

After life test: 0.175 volt.

Overload current: 80 amperes dc,
120 amperes ac.

Rupture current: 100 amperes dc,
150 amperes ac.

Duty rating: Continuous.

RFI specification: MIL-STD-461.
(applicable to coil circuits of ac operated relays)

Conformance inspection.

Group A acceptance reports shall be submitted to the qualifying activity every 36 months in order to retain qualification for this military specification sheet.

Group B and group C testing are not required. The manufacturer shall notify the qualifying activity in the event of any design or construction changes and shall impose additional testing requirements as necessary.

Qualification by similarity: See MIL-PRF-6106.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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Referenced documents. In addition to MIL-PRF-6106, this document references the following:

SAE-AS25036 MIL-STD-461

Custodians:
Navy - AS
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5945-2024-063)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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